

Cover letter — IEEE Transactions on Geoscience and Remote Sensing (TGRS)

Dear Editor-in-Chief and Associate Editors,

We submit our manuscript, “Geographic coverage debt of frozen Earth-observation foundation models: an encoder-robustness reliability diagnostic and a conditional target-recalibration remedy,” for consideration in IEEE Transactions on Geoscience and Remote Sensing.

Across thirteen frozen geospatial foundation-model encoders evaluated on GEO-Bench BigEarthNet-S2 under real cross-region shift, every encoder meets its in-distribution false-negative-rate risk-control guarantee, yet all thirteen incur a real reliability debt once deployed off-region, with false-negative rates running 2.6-4.7 times the nominal target. A deterministic accuracy boundary (about 0.80) marks where this audit stops mattering, since prediction sets collapse to singletons above it. An unlabeled covariate-shift correction recovers only part of the debt, but a labeled target-region slice (about 30% of target data) reliably restores the guarantee, at the cost of larger yet still informative prediction sets (22-24 of about 27 active labels).

This work gives practitioners finite-sample statistical guarantees on when a frozen Earth-observation foundation model can be trusted after a regional move, and a concrete labeled-data prescription for restoring that guarantee once shift breaks it — a distribution-free reliability audit, built on split-conformal risk control, that speaks directly to TGRS’s readership deploying these encoders operationally. The manuscript also reports its negative and conditional findings as plainly as its positive ones, which we regard as a strength for a reliability-focused audit.

We confirm that this is original work, that it is not under consideration elsewhere, and that the single author has approved the submission.

Thank you for your consideration.

Sincerely,

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Date: [at submission]